DDR3 / DDR4 Memory Routing

The hardest commonly-seen high-speed interface — done right

Byte-lane match • DQS strobes • Fly-by topology • ODT

Audience: Intermediate / Advanced Engineering Students

Topology • Length match • Termination • Reference voltages • Power

Why DDR Layout is Hard

Highest-speed parallel bus on most products

Speed

DDR4 = 3200 MT/s
UI = 312 ps
Little timing margin.

Many signals

DDR4 x16: 16 DQ +
2 DQS + 17 ADDR + CTRL = ~45 nets

Length matching

Within byte-lane to < 30 ps
Between byte-lanes to clock

Termination

ODT (on-die) + ATX VTT + VREF
must all align

Power integrity

Switching currents on VDDQ
drive PDN design

Reference rails

VREF must be µV-quiet
VTT must be low-Z

DDR Signal Groups

Every DDR design has these classes of signals

Signal class	Width (×8)	Purpose	Routing constraint
DQ (data)	8 per byte lane	Bidirectional data	Match to DQS within ±30 ps
DQS / DQS_b	1 diff pair / byte	Strobe + clock	Match within pair < 5 ps
DM (data mask)	1 per byte	Mask write data	Match to byte lane
ADDR (address)	~14-16 single-ended	Address	Fly-by, match to CLK
CMD / CTRL	~6	RAS/CAS/WE/RESET	Same as ADDR
CLK / CLK_b	1 diff pair	Clock signal	Master length, all others reference
VREF	Static (DC)	DQ + ADDR reference V	Quiet, dedicated trace, decoupled
VTT	Static (DC)	Termination voltage	Heavy decoupling + low Z

Topology — Fly-By vs T-Branch

How ADDR/CMD connect to multiple DRAM chips

Two topologies for shared signals

Fly-by (DDR3+ standard)

- MCU → DRAM 1 → DRAM 2 → DRAM N → VTT term
- Each DRAM gets the signal in sequence
- Lower stub loading; better SI
- Requires write-leveling to compensate for skew
- Standard for DDR3 / DDR4 / DDR5
- Termination at the far end (VTT)

T-branch (DDR2 legacy)

- MCU → fork → all DRAMs in parallel
- All DRAMs receive simultaneously
- Higher stub loading; worse SI past DDR2
- Used in older DDR2 designs
- Easier to route
- Termination per DRAM end

Byte-Lane Length Matching

DQ + DQS + DM all travel together

- Each byte lane = 8 DQ + 1 DQS pair + 1 DM = 11 nets
- Within byte lane: match all DQ to DQS within ±30 ps (≈ 5 mm in FR-4)
- DDR4 tightens to ±20 ps (~3.3 mm)
- DDR5 tightens further — read vendor app notes
- Strobe (DQS) is the timing reference — DQ + DM follow
- Match DQS diff pair within pair < 5 ps (~0.8 mm)
- Use length-tuning serpentines on shorter traces
- Distance from MCU to each byte lane: match to clock within ±50 ps

Clock + ADDR/CMD Matching

Match length to clock, not each other

- CLK / CLK_b = differential clock — master reference
- ADDR / CMD / CTRL signals: match LENGTH to CLK at the DRAM, within ±25 ps (DDR4)
- Order of arrival doesn't matter — only matching to clock matters
- Length-match across all DRAMs (fly-by: easy; T-branch: hard)
- CLK diff pair: match within pair < 5 ps
- Layer choice: route CLK + ADDR + CMD on same layer for predictable Z_0
- Reference plane: solid GND directly below — no exceptions
- Stitching vias at every layer change

Stackup for DDR

What goes on which layer

- Minimum 6-layer for DDR3; 8+ layer common for DDR4
- Typical 8-layer stackup:
- L1 (top): Signal — high-speed data / strobe
- L2: GND plane (reference for L1)
- L3: PWR plane (VDDQ + VREF + analog rails)
- L4: Signal — internal data layer
- L5: GND plane
- L6: PWR plane (VTT + control rails)
- L7: Signal — ADDR/CMD/CTRL
- L8 (bottom): Signal + clock
- Critical: every high-speed signal on a layer adjacent to GND

Termination — ODT + VTT

On-die + external — both needed

- ODT (On-Die Termination) — programmable per DRAM (40, 60, 80, 120 Ω)
- Eliminates need for many external resistors (DDR3+ standard)
- Still needs VTT termination on ADDR / CMD / CTRL signals (fly-by far end)
- VTT = VDDQ / 2 — typically 0.6 V for DDR4 (VDDQ = 1.2 V)
- VTT supply must be LOW IMPEDANCE — dedicated LDO (TPS51200 / SY79100B)
- VREF = VDDQ / 2 — clean reference for DQ and ADDR sampling
- VREF must be µV-quiet — RC filter from VDDQ; bypass cap per device
- VREF and VTT are NOT the same rail (different load demands)

VREF Generation

The most-overlooked DDR critical rail

- VREF = the reference voltage receivers compare to (for HIGH vs LOW detection)
- DDR3: VREF = 0.75 V (VDDQ = 1.5 V); DDR4: VREF = 0.6 V (VDDQ = 1.2 V)
- DDR5: per-byte VREF (more flexibility, per-lane training)
- Generation: external precision resistor divider, OR vendor IC (TPS51200, ISL68xx)
- Filter caps at every DRAM and at the MCU VREF input pin
- Common bug: shared VREF on a noisy plane → bit errors at high speed
- Star routing from VREF source to all consumers
- DDR4: VREF generated INSIDE the MCU (not externally) — but still needs clean PCB routing

Power Integrity for DDR

Where DDR designs fail most often

- VDDQ supply must deliver burst current at fast edges
- Decoupling: 10 µF + 1 µF + 100 nF + 10 nF at every DRAM power pin
- Bulk caps on VDDQ rail near MCU (47 µF) + per-byte-lane (10 µF)
- PDN (Power Distribution Network) impedance target: < 10 mΩ @ 1 GHz
- Plane capacitance: use closely-spaced VDDQ + GND planes for HF decoupling
- Inrush current on VDDQ rail can saturate the LDO/buck — check transient response
- VTT supply: sinks AND sources current → push-pull LDO required (not standard LDO)

Layer Transitions for DDR Signals

Stitching vias every time

- When a DDR signal changes layer (via), its return current must follow
- Place GND stitching vias adjacent to signal vias (within $\lambda/20$ = ~3 mm @ 1.6 GHz)
- Stitching vias provide low-impedance return path between reference planes
- Without stitching → return current detours → impedance jump + EMI
- For diff pair via transitions: both signal vias + 2-4 stitching vias nearby
- Microvias for fine-pitch BGAs — minimum via inductance
- Back-drilling reduces via stubs for very high-speed DDR4 / DDR5

DDR Design Tools

Don't design DDR by hand

Vendor design guides

Micron MT41K/MT40A
Samsung K4A8G
Hynix H5AN — must-read.

MCU vendor app notes

STM32H7 / NXP i.MX8 /
Xilinx Versal — DDR pinout + topology.

HyperLynx DDR

Cadence/Mentor SI tool
Pre-layout topology check.

Sigrity PowerSI

Cadence PI tool
PDN analysis for VDDQ rail.

KiCad

Has length-tuning + diff pair routing
Manually drive constraints.

Pre-layout sim

Run BEFORE PCB layout
Fix topology issues early.

Common DDR Mistakes

What 'random read errors' really mean

- Length mismatch > 30 ps within byte lane → flaky data above 1600 MT/s
- DQS diff pair on different layers → no CMRR, write fails
- VREF on a noisy plane → bit errors at high speed
- VTT generator not low-Z (regular LDO) → ringing, address mis-decode
- Skipping fly-by topology on DDR4 → SI failure
- Address signals routed without matching to CLK → setup violations
- Decoupling caps too far from DRAM power pins → undershoot/overshoot
- Stitching vias missing at layer changes → return path discontinuity

Best Practices

Hard rules for production DDR designs

- Read MCU + DRAM vendor design guides FIRST
- Use vendor reference design as starting point
- 6+ layer stackup with GND adjacent to high-speed
- Fly-by topology for ADDR/CMD/CLK
- Byte-lane match: DQ to DQS within 30 ps
- DQS diff pair within pair < 5 ps

- VTT dedicated low-Z source (push-pull)
- VREF: precision divider + heavy bypass
- Continuous GND under all DDR signals
- Stitching vias at layer changes
- Pre-layout SI simulation before fab
- Test with memtest86 + traffic generator

Key Takeaways

Key takeaways from this lecture

Byte-lane match

DQ to DQS within 30 ps (DDR4).

Fly-by for ADDR/CLK

DDR3+ default topology.

ODT + VTT

Both needed for clean signals.

VREF is sacred

µV-quiet, dedicated rail.

PDN matters

VDDQ rail Z < 10 mΩ @ 1 GHz.

Simulate first

HyperLynx / Sigrity before layout.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- Micron / Hynix / Samsung DDR layout design guides
- Intel / NXP / Xilinx MCU+DDR reference designs
- JEDEC DDR3/4/5 specifications
- HyperLynx DDR / Sigrity PowerSI — commercial SI/PI tools
- Eric Bogatin — DDR-specific chapters
- Robert Feranec YouTube — DDR4 layout walkthrough
- STM32H7 / RT1170 reference design schematics + Gerbers
- PCB West / DesignCon — annual SI/PI conferences

Thank you • Build something this week.